

Problem 1:

We call n be the size of initial problems
so each time we split, we get:

$$(1-\sigma) \cdot n \text{ and } \sigma \cdot n$$

Minimum depth appears when we go follow σn
branch because size decrease very fast

$$n, \sigma n, \sigma^2 n, \dots, \sigma^k \cdot n$$

$$\text{leaf} \Rightarrow \sigma^k \cdot n \leq 1 \quad (1)$$

$$\Rightarrow \sigma^k \leq \frac{1}{n}$$

$$\Rightarrow \log(\sigma^k) \leq \log\left(\frac{1}{n}\right)$$

$$\Rightarrow k \cdot \log(\sigma) \leq -1 \cdot \log(n)$$

$$\Rightarrow k \geq \frac{-1 \log(n)}{\log(\sigma)}$$

because $\log(\sigma) < 0$ when $0 < \sigma < \frac{1}{2}$

$$\text{so min depth is } \left\lceil \frac{-1 \cdot \log(n)}{\log(\sigma)} \right\rceil$$

maximum depth:

we apply to (1) then

$$(1-\sigma)^k \cdot n \leq 1$$

$$\Rightarrow (1-\sigma)^k \leq \frac{1}{n}$$

$$\Rightarrow k \cdot \log(1-\sigma) \leq \log\left(\frac{1}{n}\right)$$

$$\Rightarrow k \geq \frac{-1 \cdot \log(n)}{\log(1-\sigma)}$$

So max depth is

$$\left\lceil \frac{-1 \cdot \log(n)}{\log(1-\sigma)} \right\rceil$$

Problem 2:

Pseudo code:

MIN-HEAPIFY(A, i)

l = 2*i

r = 2*i + 1

min = i

if l <= heap_size(A) and A[l] < A[min]

min = l

if r <= heap_size(A) and A[r] < A[min]

min = r

if min != i

swap A[i] and A[min]

MIN-HEAPIFY(A, min)

In worst case, we might have to fix up to the height of the tree, we know $h \approx O(\log n)$

So $T(n) = O(\log n)$ * some operations take constant time

In general, worst case is $O(\log n)$.

Problem 3:

COW,DOG,SEA,RUG,ROW,MOW.BOX,TAB,BAR,EAR,TAR,DIG,BIG,TEA, NOW,FOX.

Rearrange:

COW

DOG

SEA

RUG

ROW

MOW

BOX

TAB

BAR

EAR

TAR

DIG

BIG

TEA

NOW

FOX

We start from the last letter:

A: SEA, TEA

B: TAB

G: DOG, RUG, DIG, BIG

R: BAR, EAR, TAR

W: COW, ROW, MOW, NOW

X: BOX, FOX

After this we have:

SEA, TEA, TAB, DOG, RUG, DIG, BIG, BAR, EAR, TAR, COW, ROW, MOW, NOW,
BOX, FOX

We start the middle letter:

A: TAB, BAR, EAR, TAR

E: SEA, TEA

I: DIG, BIG

O: DOG, COW, ROW, MOW, NOW, BOX, FOX

U: RUG

After this we have:

TAB, BAR, EAR, TAR, SEA, TEA, DIG, BIG, DOG, COW, ROW, MOW, NOW, BOX,
FOX, RUG

The 1st letter:

B: BAR, BIG, BOX

C: COW

D: DIG, DOG

E: EAR

F: FOX

M: MOW

N: NOW

R: ROW, RUG

S: SEA

T: TAB, TAR, TEA

After this we have:

**BAR, BIG, BOX, COW, DIG, DOG, EAR, FOX, MOW, NOW, ROW, RUG, SEA, TAB,
TAR, TEA**

Problem 4:

$$a) T(n) = 2T(n/4) + 1$$

$$a = 2, b = 4, f(n) = 1$$

$$\log_4 2 = \frac{1}{2}$$

we observe:

$$f(n) = O(1) = O(n^{\frac{1}{2} - \epsilon})$$

$$\text{with } \epsilon = \frac{1}{2}$$

$$\Rightarrow T(n) = \Theta(n^{\frac{1}{2}}) = \Theta(\sqrt{n})$$

$$b) T(n) = 2T(n/4) + \sqrt{n}$$

$$= 2T(n/4) + n^{\frac{1}{2}}$$

$$n^{\log_4 2} = n^{\log_4 2} = n^{\frac{1}{2}}$$

We see

$$f(n) = \sqrt{n} = \Theta(n^{\frac{1}{2}})$$

$$\begin{aligned} \Rightarrow T(n) &= \Theta(n^{\frac{1}{2}} \cdot \log n) \\ &= \Theta(\sqrt{n} \cdot \log n) \end{aligned}$$

$$c) \quad T(n) = 2T\left(\frac{n}{4}\right) + n$$

$$n^{\log_4 a} = n^{\log_4 2} = n^{\frac{1}{2}}$$

we see

$$f(n) = n = \Omega(n) = \Omega(n^{\frac{1}{2} + \epsilon})$$

$$\Rightarrow \epsilon = \frac{1}{2} \text{ and:}$$

$$a \cdot f\left(\frac{n}{4}\right) = 2 \cdot \frac{n}{4} = \frac{n}{2} < c \cdot n$$

We choose $c = 0.75$

$\Rightarrow$ Master theorem valid

$$\Rightarrow T(n) = \Theta(n)$$

$$d) \quad T(n) = 2T\left(\frac{n}{4}\right) + n^2$$

$$f(n) = n^2$$

$$n^{\log_4 a} = n^{\frac{1}{2}}$$

$$\text{we see: } f(n) = n^2 = \Omega(n^{\frac{1}{2} + \epsilon})$$

here $\epsilon = \frac{3}{2}$ and

$$a \cdot f\left(\frac{n}{4}\right) = 2 \cdot \frac{n^2}{4^2} = \frac{n^2}{8}$$

$$\text{to have } \frac{n^2}{8} < c \cdot n^2$$

we choose $c = \frac{1}{2} \Rightarrow$ master theorem valid

$$\Rightarrow T(n) = \Theta(n^2)$$